
MACROMOLECULAR COMPOUNDS
AND POLYMERIC MATERIALS

Synergism of Triallyl Cyanurate and Divinylbenzene in a Dynamically Vulcanized Blend of Polypropylene with Hydrogenated Styrene–Butadiene–Styrene Block Copolymer

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Abstract—The synergistic effect of divinylbenzene and triallyl cyanurate on the properties and structure of the thermoplastic vulcanizate prepared from a blend of polypropylene and hydrogenated styrene–butadiene–styrene triblock copolymer by vulcanization under dynamic conditions with 2,5-dimethyl-2,5-di(*tert*-butylperoxy)hexane was revealed. The synergism consists in the enhancement of the strength, increase in the relative elongation, and decrease in the compression set to a greater extent than when using divinylbenzene or triallyl cyanurate separately. The synergism is caused by the increased gel fraction content and vulcanization network thickness, with preservation of the higher molecular mass of the polypropylene matrix.

Keywords: hydrogenated styrene–butadiene–styrene block copolymer, thermoplastic vulcanizate, residual strain, divinylbenzene, triallyl cyanurate

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The production of thermoplastic elastomers undergoes active development. Thermoplastic elastomers based on hydrogenated styrene–butadiene–styrene block copolymer (HSBS) and their blends with other polymers exhibit high mechanical characteristics and resistance to aging, UV radiation, and low temperatures; therefore, they have found the widest use today [1–4]. However, these thermoplastic elastomers have a considerable drawback compared to classical vulcanized rubbers and dynamically vulcanized blends of polyolefins with elastomers: increased residual strain [4], which considerably narrows their application field. Therefore, elimination of this drawback is a topical problem.

A classical way to reduce the residual strain of elastomers and thermoelastoplastics based on blends of plastics with elastomers is their vulcanization. Thermoplastic elastomers with the vulcanized elastomer phase are usually termed thermoplastic vulcanizate [5]. In thermoplastic vulcanizates the elastomer phase is chemically cross-linked directly in the course of

blending. In HSBS, there are no double bonds or other chemical groups ensuring the vulcanization ability via various routes. Therefore, the only suitable pathway of HSBS vulcanization is the free-radical pathway, e.g., under the action of peroxides or ionizing radiation. Introduction of peroxide [6] negatively influences the residual strain and the strength properties of polypropylene–HSBS blends, which is caused by strong degradation of polypropylene macromolecules and very weak degree of HSBS cross-linking [6]. As shown previously [6, 7], the residual strain of thermoplastic elastomer based on propylene–HSBS blend can be reduced by peroxide vulcanization of the elastomer phase in the presence of coagents. We studied a series of peroxide vulcanization coagents such as triethylene glycol trimethylolpropane, styrene–butadiene–styrene block copolymer [7], triallyl cyanurate, and triethylene glycol dimethacrylate [6], of which triallyl cyanurate showed the highest performance. However, because of degradation of polypropylene macromolecules in the course of peroxide vulcanization, the strain and strength

characteristics of the thermoplastic elastomer appear to be unsatisfactory [6, 7].

On the other hand, it is known from the literature that divinylbenzene is a peroxide vulcanization coagent that efficiently prevents the polypropylene degradation and leads to the formation of a certain amount of the gel fraction in it [8]. In this connection, it was interesting to study the effect of divinylbenzene as a coagent of peroxide dynamic vulcanization of a blend of polypropylene with HSBS and to compare it with the coagent that previously showed the best results, triallyl cyanurate.

EXPERIMENTAL

To prepare thermoplastic vulcanizate based on HSBS, we used polypropylene of PP8300G grade. It is a block copolymer of propylene with a small amount of ethylene. Its characteristics are as follows: melt flow index (MFI) 1.2 g/10 min, bending elastic modulus (E_b) 1300 MPa, and viscosity-average molecular mass (MM) 125 000 g mol⁻¹. To prepare thermoplastic vulcanizate based on ethylene-propylene rubber (EPDM), we used polypropylene of PP1500J grade (isotactic homopolypropylene) with the following characteristics: MFI = 3.1 g/10 min and E_b = 1400 MPa. Both kinds of polypropylene are produced by PAO Nizhnekamskneftekhim. As HSBS we chose Septon V9461 product (Kurarey, Japan) containing 30 wt % bound styrene, with the viscosity of a 5% solution in toluene equal to 0.09 Pa s at 25°C. As EPDM we used rubber of Vistalon 5601 grade (ExxonMobil), containing 69 wt % ethylene and 5 wt % ethyldenenorbornene, with the Mooney viscosity of 72 [at mL (1 + 4) 125°C]. Model studies were performed using polystyrene of PS525 grade (Nizhnekamskneftekhim) with MFI = 9 g/10 min (at 200°C and a load of 5 kg).

White paraffin oil of Primol 382 grade (ExxonMobil) with the kinematic viscosity of 70 mm² s⁻¹ at 40°C was used as a plasticizer for both types of thermoplastic vulcanizates.

Vulcanization of a blend of polypropylene with HSBS was performed with a peroxide, 2,5-dimethyl-2,5-di(*tert*-butylperoxy)hexane (DTBPH, trade name Trigonox 101, AkzoNobel). Triallyl cyanurate (Kargin Research Institute of Polymers) and divinylbenzene (Merck, mixture of isomers, boiling point 195°C) were used as peroxide vulcanization coagents.

Experiments were performed with a blend with a constant polypropylene : HSBS : oil weight ratio equal to 30 : 70 : 70 and varied content of the peroxide and vulcanization coagents.

Vulcanization of a polypropylene-EPDM blend was performed using SP-1045 resol phenol-formaldehyde resin (SI Group) containing 8.6 wt % methylol groups. SnCl₂·2H₂O served as a vulcanization activator. The thermoplastic vulcanizate based on polypropylene and EPDM contained the following components (wt parts): polypropylene/EPDM/oil/SP-1045/SnCl₂·2H₂O = 30/70/70/3/0.75.

The thermoplastic vulcanizates that we prepared contained an antioxidant (0.9% Irganox 1010) and a stabilizer (0.2% Irgafos 168, both BASF).

The compounding and vulcanization were performed at 170°C with a Brabender laboratory mixer with the chamber volume of 60 mL at a rotor rotation rate of 150 rpm. First we compounded polypropylene with the elastomer and oil for 6 min, after which we introduced the vulcanizing agents over a period of 1 min and continued compounding for 3 min to complete the vulcanization.

The nominal tensile strength (σ) and relative elongation at break (ϵ) were measured according to ISO 37:2011 using type 1 blades at a clamp velocity of 500 mm min⁻¹. The blades were cut from 2-mm-thick plates obtained by pressing at 180°C with keeping under pressure for 5 min. The elongation set (ϵ_{set}) was measured under the same conditions according to GOST (State Standard) 270-75, Method for Determining the Tensile Elastic and Strength Characteristics.

The Shore A hardness was measured according to GOST R ISO 7619-1-2009 with an IT5078 hardness meter at 23 ± 2°C.

The static compression set (SCS) was determined according to GOST R ISO 815-1-2017, Rubber and Thermoplastics. Determination of the Compression Set, using method A at 25% compression and two different modes of keeping in the compressed state: 22 h at 70°C or 500 h at 125°C. Cylindrical specimens 13 mm in diameter and 6 mm high for SCS measurements were prepared by pressing at 180°C with keeping under pressure for 5 min.

Sol-gel analysis of the thermoplastic vulcanizate was performed by extraction of the sol fraction with refluxing *o*-xylene in a Soxhlet apparatus [9]. The undissolved residue was dried for 80°C for 24 h, and the

dry residue was considered as the gel fraction (GF, %). The relative content of the gel fraction was calculated using the formula

$$\text{GF} = \frac{m_1}{m_0} \times 100,$$

where m_1 is the weight of the dried gel fraction, and m_0 is the sample weight before the extraction.

The molecular mass of polypropylene, initial and extracted from the thermoplastic vulcanizate, was determined viscometrically with an Ubbelohde viscometer at 135°C in a decalin solution. The molecular mass was calculated by the Mark–Houwink equation

$$[\eta] = K \times \text{MM}^\mu,$$

where $[\eta]$ is the intrinsic viscosity (mL g^{-1}), and K and μ are constants (for the polypropylene–decalin system at 135°C, 0.001580 and 0.77, respectively).

The density v of the vulcanization network of the gel fraction isolated from the thermoplastic vulcanizate was determined from the equilibrium swelling in toluene. To this end, the dry gel fraction was placed in toluene and allowed to stand at $23 \pm 2^\circ\text{C}$ for 24 h. The vulcanization network density was calculated using the Flory–Rehner equation [10]

$$v = \frac{\rho_2}{\mu_2} \frac{\ln(1 - \phi_1) + \phi_1 + \chi \phi_1^2}{\phi_1^{1/3} - \frac{2}{f} \phi_1},$$

where ρ_2 is the solvent density (g cm^{-3}); μ_2 , molecular mass of the solvent (g mol^{-1}); f , functionality of the network chains ($f = 4$); χ , polymer–solvent interaction parameter (0.503 at 20°C) [11]; and ϕ_1 , volume fraction of the elastomer in the swollen sample,

$$\phi_1 = \frac{m_3 / 0.91}{m_3 / 0.91 + (m_3 - m_2) / \rho_2},$$

where m_2 is the weight of the dry GF sample before immersing in toluene (g), m_3 is the gel fraction weight after swelling in toluene (g), and 0.91 g cm^{-3} is the HSBS density.

The oil resistance of the thermoplastic vulcanizate (α) was determined from the degree of swelling of the sample in accordance with GOST R 1817–2016, Rubber and Thermoelastoplastics. Determination of the Resistance to the Action of Liquids, method A, in oil no. 3. Plates 2 mm thick were kept in oil for 168 h

at 100°C, after which the relative weight gain of the sample was determined.

The intensity of the vulcanization occurring on compounding was evaluated by two parameters: rate of an increase in the torque on the mixer shaft (V_m) and torque increment in the course of vulcanization (ΔN). In the curve of the time dependence of the torque, we chose the portion corresponding to an increase in the torque in the course of vulcanization and calculated V_m as the ratio of the torque increment to the time in which the torque increased. The quantity ΔN was calculated from the time dependence of the torque as the difference between the torque in the peak formed in the course of dynamic vulcanization to the torque before the vulcanization start.

The degree of crystallinity (C_{cr}) of polypropylene in the thermoplastic vulcanizate was determined by differential scanning calorimetry [12] with a Q-200TA device in a nitrogen atmosphere at a heating rate of 2 deg min^{-1} . Prior to measurements, the thermoplastic vulcanizate samples were annealed at 130°C for 1 h to level off differences in the crystal structure of the samples, caused by the effect of the preparation temperature and time. When calculating the degree of crystallinity, we took into account the fraction of polypropylene in the thermoplastic vulcanizate. As the thermal flux in melting of 100% crystalline polypropylene, we used the value of 160 J g^{-1} [12].

The melt flow index was determined with an IIRT-type capillary viscometer in accordance with GOST 11645–73, Plastics. Method for Determining the Melt Flow Index of Thermoplastics. Measurement conditions: for thermoplastic vulcanizate, 190°C, 10 kg load; for polystyrene, 200°C, 5 kg load.

RESULTS AND DISCUSSION

The experiment has shown that the addition of divinylbenzene or triallyl cyanurate into a blend of polypropylene with HSBS and oil allows a significant increase in the gel fraction content of the thermoplastic vulcanizate and in the thickness of the vulcanization network in the gel fraction, and also the protection of the polypropylene from degradation (Table 1). The higher is the concentration of the coagents, the higher are the gel fraction content, the vulcanization network thickness, and the molecular mass of polypropylene in the thermoplastic vulcanizate. As a result, the SCS

Table 1. Properties of thermoplastic vulcanizate based on a blend of polypropylene, hydrogenated styrene–butadiene–styrene block copolymer, and oil with different content of vulcanization coagents and 1 wt % di(*tert*-butylperoxy)hexane

Compound, wt %	Gel fraction content, wt %	Vulcanization network density $\nu \times 10^4$	Polypropylene molecular mass, g mol ⁻¹	Compression set, ^a %	Melt flow index, g/10 min	Tensile strength, MPa	Elongation at break ϵ , %	Torque increment, N m	Torque increase rate, N m s ⁻¹
Divinylbenzene:									
0	11	0.1	64800	64	79	4.4	590	0	0
0.5	37	0.15	73000	35	15	6.3	500	5.9	0.9
1	40	0.19	82200	37	1	6.3	385	7.1	1.04
2	44	0.32	98000	40	0.5	6.1	330	7.3	1
Triallyl cyanurate:									
0.5	37	0.23	67000	42	24	4.5	340	1.5	0.2
1	40	0.31	69580	37	2.5	4.5	330	2	0.26
2	41	0.49	80000	38	2	4.6	300	3	0.54

^a SCS measurement conditions: 70°C, 22 h.

and MFI of the thermoplastic vulcanizate decrease and its strength increases relative to the thermoplastic vulcanizate obtained with DTBPH only.

Comparison of divinylbenzene and triallyl cyanurate shows that divinylbenzene exhibits higher vulcanization performance, as seen from the rate of an increase in the torque on the mixer shaft and from the absolute increment of the torque on the shaft (Table 1). In addition, divinylbenzene protects polypropylene from degradation more efficiently, as seen from the higher molecular mass of polypropylene in the thermoplastic elastomer (Table 1). On the other hand, triallyl cyanurate gives higher thickness of the vulcanization network in the gel fraction. However, despite such differences, both coagents increase the GF content of the thermoplastic vulcanizate and decrease SCS of the polymer blend to an approximately equal extent (Table 1). However, divinylbenzene allows reaching higher values of σ and ϵ , which is most probably due to higher molecular mass of the polypropylene matrix (Table 1).

Higher strength of the thermoplastic vulcanizate compared to divinylbenzene may also be due to other factors. Peroxide causes active degradation of the polystyrene macromolecule [13]. Such degradation cannot be excluded also in polystyrene blocks of the HSBS macromolecule in the course of the peroxide vulcanization. Presumably, the introduced divinylbenzene protects the polystyrene block from the

degradation more efficiently than triallyl cyanurate does. It was interesting to check this assumption. However, it is impossible to cut out the polystyrene block from the thermoplastic vulcanizate to measure its molecular mass. Therefore, we performed model studies of the effect of the peroxide and vulcanization coagents on polystyrene and, instead of the polystyrene block, took polystyrene separately. Introduction of 1% DTBPH into polystyrene led to an increase in its MFI from 9 to 70 g/10 min (measurement at 200°C and 5 kg). Such an increase in MFI of polystyrene can be attributed only to a decrease in its molecular mass. Introduction of 1% divinylbenzene in addition to 1% DTBPH allows MFI of polystyrene to be preserved on the level of 15 g/10 min, and replacement of divinylbenzene in polystyrene by triallyl cyanurate gives the MFI as high as 65 g/10 min. Hence, divinylbenzene protects polystyrene from degradation under the action of peroxy radicals considerably more efficiently than triallyl cyanurate does. This may be one more cause of higher strength of the thermoplastic elastomer vulcanized with DTBPH in the presence of divinylbenzene. It should be noted that introduction of DTBPH and coagents into polystyrene does not cause the formation of a gel fraction, even in a small amount.

Because divinylbenzene protects the polypropylene and polystyrene macromolecules from degradation more efficiently and triallyl cyanurate allows reaching

Table 2. Variation of the parameters of polymer components in the thermoplastic vulcanizate based on a blend of polypropylene, hydrogenated styrene–butadiene–styrene block copolymer, and oil with variation of the divinylbenzene/triallyl cyanurate ratio in the presence of 1% di(*tert*-butylperoxy)hexane

Triallyl cyanurate, wt %	Divinylbenzene, wt %	Gel fraction content, wt %	Torque increase rate, N m s ⁻¹	Torque increment, N m	Polypropylene molecular mass, g mol ⁻¹	Vulcanization network density $\nu \times 10^4$, mol mL ⁻¹	Degree of crystallinity C_{cr} , %
0	0	11	0	0	64800	0.1	45
0	1	40	1.04	7.1	82200	0.19	49
0.2	0.8	42	0.50	7.0	83000	0.23	49
0.6	0.4	48	0.39	6.4	81000	0.46	49
0.8	0.2	43	0.30	5.0	72000	0.52	48
1	0	40	0.26	2.0	69580	0.31	46

higher thickness of the vulcanization network (Table 1), enhanced vulcanization efficiency in the case of joint use of divinylbenzene and triallyl cyanurate can be expected.

Figure 1 shows that variation of the triallyl cyanurate/divinylbenzene ratio at their constant total concentration leads to a nonadditive change in the majority of the elastic and strength characteristics of the thermoplastic

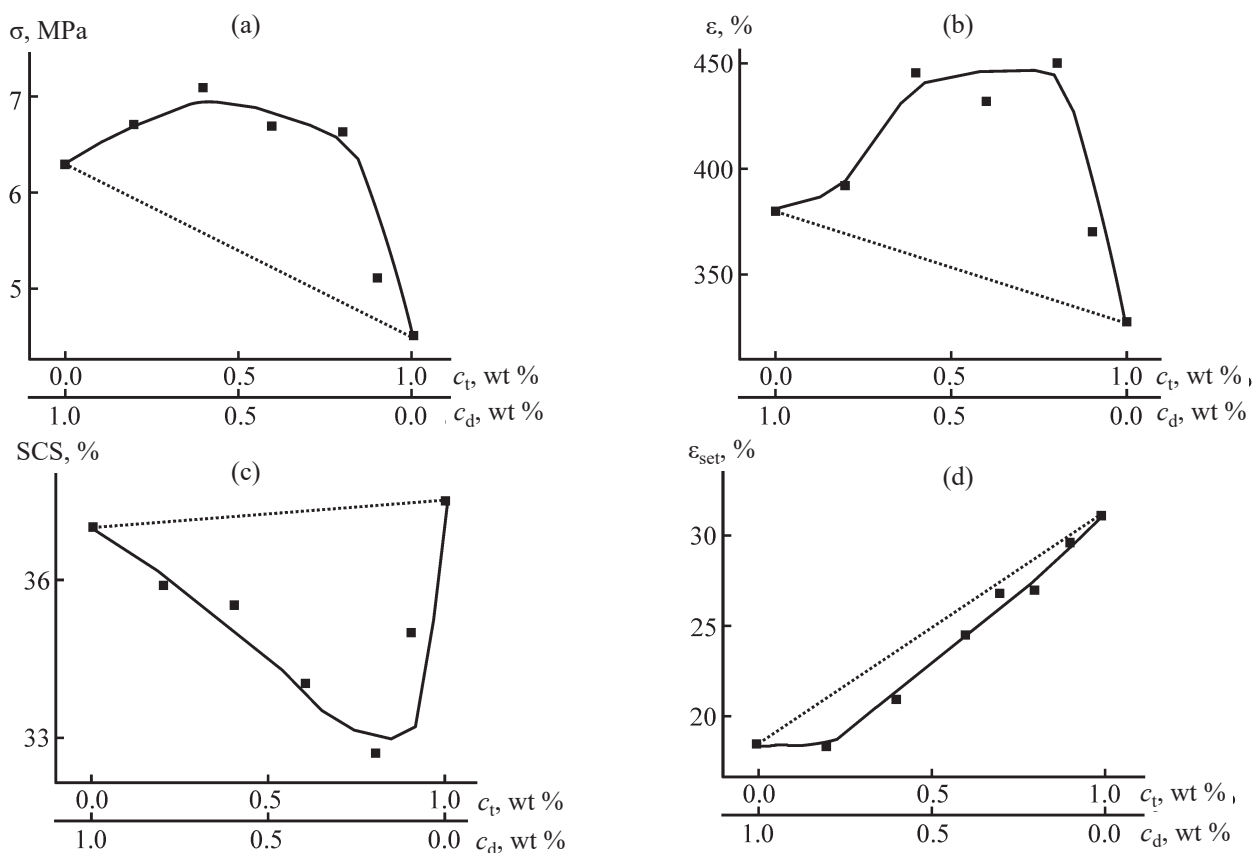


Fig. 1. Plots of (a) σ , (b) ϵ , (c) SCS (70°C, 22 h), and (d) ϵ_{set} of the thermoplastic vulcanizate based on HSBS, polypropylene, and oil vs. (c_t) triallyl cyanurate to (c_d) divinylbenzene ratio. Total content of coagents 1 wt %, DTBPH content 1 wt %.

Table 3. Properties of thermoplastic vulcanizates based on the polypropylene/hydrogenated styrene–butadiene–styrene block copolymer/oil/dimethyldi(*tert*-butylperoxy)hexane/triallyl cyanurate/divinylbenzene blend (30/70/70/1/0.8/0.2 wt parts) and on the polypropylene/ethylene–propylene rubber/oil blend. Hardness of both thermoplastic vulcanizates 65 Shore A units

Base of thermoplastic vulcanizate	Compression set, ^a %	Compression set, ^b %	Breaking tensile stress, MPa	Relative elongation at break, %	Swelling in oil, %
Polypropylene and hydrogenated styrene–butadiene–styrene block copolymer	32	44	7.25	480	93
Polypropylene and ethylene–propylene rubber	39	85	6.2	380	95

^a After keeping for 22 h at 70°C.

^b After keeping for 500 h at 125°C.

vulcanizate. The deviations from the additivity are very significant and positive; they improve the operation characteristics of the thermoplastic vulcanizate. For example, σ and ε (Figs. 1a, 1b) show positive deviations, and SCS, negative deviation (Figs. 1c, 1d). This allows preparation of the thermoplastic vulcanizate with higher consumer's properties. The observed nonadditive changes allow us to speak of the synergistic effect of divinylbenzene and triallyl cyanurate on the properties of the peroxide-vulcanized polypropylene–HSBS blend.

What are the causes of such synergistic effect of the coagents? The polypropylene–HSBS system is incompatible [1, 3]. The thermoplastic vulcanizate based on it consists of four phases: polypropylene (crystalline and amorphous phases), polystyrene, and ethylene–propylene random copolymer formed in the course of hydrogenation of the polybutadiene fragment of the styrene–butadiene–styrene block copolymer macromolecule [4]. To account for the nonadditive changes in the properties of the thermoplastic vulcanizate, it is necessary to examine the effect of triallyl cyanurate and divinylbenzene on each phase.

Our experiments have shown that the molecular mass, degree of crystallinity of polypropylene, ΔN , and V_m vary approximately additively with variation of the divinylbenzene/triallyl cyanurate ratio (Table 2). Joint introduction of divinylbenzene and triallyl cyanurate leads to higher gel fraction content and increased thickness of the vulcanization network in the gel fraction, compared to the introduction of the coagents separately in the same amount (Table 2). It is commonly known that an increase in the gel fraction content and in the

vulcanization network thickness leads to a decrease in SCS and in the relative elongation and to an increase in the rubber strength. Therefore, the observed decreased values of SCS and increased values of σ of thermoplastic vulcanizates with joint introduction of triallyl cyanurate and divinylbenzene can be caused by higher values of the gel fraction content and vulcanization network thickness. However, there may be also other factor responsible for the synergistic effect with extrema in the deformation and strength characteristics.

In joint introduction of divinylbenzene and triallyl cyanurate, the gel fraction concentration in the thermoplastic vulcanizate exceeds the concentration of HSBS and vulcanizing agents taken together, which should be about 40 wt % (Table 2). Hence, in this case not only HSBS but also polypropylene participates in the gel fraction formation. Polypropylene can both undergo grafting to HSBS macromolecules and form a certain amount of the intrinsic gel fraction. Grafting of polypropylene to HSBS can enhance the adhesion between these components and make a certain contribution to enhancement of the strength of the thermoplastic vulcanizate upon joint introduction of the two coagents. The relatively high molecular mass of polypropylene, preserved in this case (Table 2), also favors an increase in σ and ε of thermoplastic vulcanizates (Fig. 1).

It was interesting to compare the properties of the thermoplastic vulcanizate based on the polypropylene–HSBS blend and the most widely used thermoplastic vulcanizate based on the blend of polypropylene with ethylene–propylene rubber [5]. The thermoplastic

vulcanizate based on the polypropylene–HSBS blend is characterized by lower SCS and higher values of σ and ϵ , compared to the thermoplastic vulcanizate based on the polypropylene–EPDM blend at equal polypropylene/elastomer ratio and equal hardness (Table 3). The advantage of the thermoplastic vulcanizate based on the polypropylene–HSBS blend is manifested most strongly in SCS at elevated temperatures and long time of keeping in the compressed state (Table 3). Both thermoplastic vulcanizates have equal moderately high resistance to oil.

A strong decrease in SCS with the preservation of the high strength of the thermoplastic vulcanizate based on the HSBS–polypropylene blend makes it possible to prolong the operation life of items that are produced today from the thermoplastic vulcanizate based on the polypropylene–EPDM blend.

CONCLUSIONS

The effect of the peroxide vulcanization coagents, triallyl cyanurate and divinylbenzene, and of their combinations on the molecular structure and properties of the thermoplastic vulcanizate based on the polypropylene–HSBS blend vulcanized with DTBPH under the conditions of compounding was studied. Both triallyl cyanurate and divinylbenzene increase the vulcanization network thickness and the gel fraction content of the thermoplastic vulcanizate and protect polypropylene and HSBS polystyrene blocks from degradation under the action of radicals. Joint introduction of triallyl cyanurate and divinylbenzene intensifies the vulcanization of the blend, increases the gel fraction content and vulcanization network thickness, and decreases SCS of the thermoplastic vulcanizate to a considerably greater extent than the triallyl cyanurate or divinylbenzene taken separately does. The synergistic effect of triallyl cyanurate and divinylbenzene is observed. This allows preparation of the thermoplastic vulcanizate with enhanced operation characteristics.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

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